

1 The Linear Gaussian Information Filter (IF)

Modifies the classical KF to address numerical issues with inverses and posdef requirements. Define $I_{k+1}^- = [P_{k+1}^-]^{-1}$ Predicted information matrix.

$I_{k+1}^+ = [P_{k+1}^+]^{-1}$ -> Updated information matrix.

Helps primarily when p (measurements) $> n$. Won't need to invert innovation matrix. Can also initialize with no information.

Caveats: Need to reconstruct the actual state estimate.

$$\hat{x}_{k+1}^+ = [I_{k+1}^+]^{-1} \hat{i}_{k+1}^+$$

Requires I to be invertible, this means the system must be observable. If you need 3 measurements for the system to be observable, you won't be able to reconstruct a state estimate until you observe three measurements.

1.1 When to use IF vs KF

IF more computationally efficient if $p > n$ and if R_k is diagonal. (More sensors than states).